

Markscheme

Mathematics: analysis and approaches

Practice question bank

71. (a)

Three consecutive terms of a geometric sequence are $x - 1$, 8, and $x + 11$, where all three terms are positive.

Find the value of x .

the square of the middle term equals the product of its neighbours: $8^2 = (x - 1)(x + 11)$ *(M1)*

$$x^2 + 10x - 75 = 0 \quad \text{A1}$$

$$(x - 5)(x + 15) = 0 \Rightarrow x = 5 \text{ or } x = -15 \quad \text{M1}$$

reject $x = -15$ since the terms must be positive; $x = 5$ **R1**

[4 marks]

(b)

Hence find the common ratio of the sequence.

the terms are 4, 8, 16 *(M1)*

$$r = 2 \quad \text{A1}$$

[2 marks]